

TriSense Capsule: A Portable Wireless Module for Urine-Based Disease Detection

S Jegadeesan

Department of Electronics and Communication Engineering
M. Kumarasamy College of Engineering
Karur - 639113, India
jegadeesans@rediffmail.com

S Suguna

Department of Electronics and Communication Engineering
M. Kumarasamy College of Engineering
Karur - 639113, India
sugunasubramani05@gmail.com

S Yogadharani

Department of Electronics and Communication Engineering
M. Kumarasamy College of Engineering
Karur - 639113, India
s.yogadharani@gmail.com

Arun Sekar Rajasekaran

Department of Electronics and Communication Engineering
SR University
Warangal, Telangana 506371, India
rarunsekar007@gmail.com

V Visnupriya

Department of Electronics and Communication Engineering
M. Kumarasamy College of Engineering
Karur - 639113, India
priya08082006@gmail.com

E Yuvasri

Department of Electronics and Communication Engineering
M. Kumarasamy College of Engineering
Karur - 639113, India
yuvasrisiva2005@gmail.com

Abstract—Affordable and reliable diagnostic facilities are one of the largest challenges of rural areas and areas that have limited resources. This paper will explain the Trisense Capsule, which is a portable urine diagnostic device that incorporates the use of electrochemical sensing in the offline mode. The capsule is powered by a low power ESP32 microcontroller, LMP91000 potentiostat and LIR2450 coin cell battery. It identifies glucose, albumin, and zinc biomarkers with the help of screen-printed electrodes to identify glucose, anti-albumin antibodies, and zinc oxide, respectively. The model was designed and simulated with the LTspice to test the behavior of the amplification, stability as well as sensitivity of the detector to signals under controlled conditions. Theoretical sensor response analyses, signal-to-noise ratio (SNR), and limit of detection (LOD) prove the viability of the system. My future activities will be in designing a prototype and confirming these performance indicators in the laboratory on synthetic urine samples. The Trisense Capsule provides good prospects of decentralized and sustainable healthcare diagnostics.

Index Terms—Electrochemical biosensing, Urinary Biomarkers, Wireless Data Transmission, Low-Power Portable Diagnostics

I. INTRODUCTION

The increasing need of cheap, mobile and decentralized healthcare diagnostics has prompted the exploration of smart biomedical sensing technologies. Traditional laboratory-based urine analysis has necessitated trained personnel, costly reagents, and central systems so that prompt identification of diseases such as diabetes, kidney diseases, urinary tract infections, and metabolic imbalances is not possible in rural or resource-strained areas [1], [3]. Specifically, the late preliminary diagnosis frequently results in late-stage treatment,

which raises the cost of health care and limits the prognosis of the patient. To overcome such limitations, scientists have been working on electrochemical and optical biosensors with low-cost to enable the detection of urine sample analytes including glucose, creatinine, uric acid, and inflammatory biomarkers [2], [4], [13].

The sensitivity and miniaturization of biosensing devices has been greatly enhanced by recent achievements in the screen-printed electrodes, nanocomposite coating, and microfluidic integration [1], [4], [7]. POC biosensors utilizing smartphones have additionally allowed real-time results without access to external laboratories, even though numerous designs still use disposable strips or stable internet connectivity to transmit data [7], [8]. Our proposal will overcome those limitations by introducing a capsule reusable biosensor, Trisense Capsule, that incorporates a set of sensors into a small chip to detect glucose, uric acid, creatinine, and other analytes in urine samples directly. The gadget is offline and sends its findings protected to a nearby smartphone through a short-range wireless communication and makes it accessible even in remote locations [9], [10].

Compared to traditional portable kits, the capsule uses low-power potentiostat circuits [14], [15] and superior sensing layers to realize consistent performance in small form factors. Besides, WBAN integration can be blockchain-enabled and fog-computing-aware to provide confidentiality and integrity to patient health records [9], [10]. These technological aspects make the Trisense Capsule a low-carbon, sustainable healthcare technology by cutting down the reliance to centralized

labs and decreasing single-use disposable diagnostics. The potential of such compact, environmentally-friendly biosensing platforms has been highly supported by new advancements in nanomaterial-based sensors [1], [2], [16] and energy-saving wireless modules [17].

II. LITERATURE REVIEW

Urine diagnostics is shifting to a new paradigm in which laboratory processes that were traditionally centralized and consumptive of reagents are being developed by portable and multiplexed point-of-care (POC) sensors that make rapid and reliable decisions near the patient. Recent experiments have also shown that nanostructured transducers and read-out systems can readily detect important urinary biomarkers including glucose, creatinine, uric acid, and disease-specific panels at clinically relevant limits at a lower cost and with minimal external infrastructure dependence [1], [13]. Specifically, photonic-crystal fiber and surface plasmon resonance (SPR) sensors, together with miniature electrochemical systems, have made it possible to have optics-free and low-power configurations, equally applicable at homes and the field [1], [3], [13].

These have resulted in major advancements in terms of innovations in electrode materials and surface recognition layers. Screen-printed electrodes in graphene and MoS₂-functionalized paper sensors increase sensitive areas of active surfaces, transfer of electrons, and sensitivity to detect glucose and lactate electrochemical reactions [1], [2]. Moreover, molecularly imprinted and multilayer nanocomposite electrodes have not only increased selectivity of low-abundance biomarkers, including troponin-T in urine, but also offer great anti-fouling properties and long-term stability [4]. Other optical biosensing applications that could be beneficial in urine sensing are complementary optical biosensing studies intended to detect heavy metal in milk, which could provide useful calibration and matrix-effect mitigation methods [11].

Recent developments in multiplexed microfluidic structures have also made it possible to integrate many zones of analytes on a single chip and remove single-strip analysis. Experiments on single-chip arrays of urine biomarkers have demonstrated differences in signal measurements and cross-talk rejection of urological diseases [3]. Microfluidic POC systems based on hybrid paper/polymer technology enable analysis, onboard fluid control, and smartphone-aided analysis, enhancing ease of use and minimizing peripheral equipment requirements [7]. The need to synchronize assays, reuse reagents, and correct drift has also been highlighted as very important in the studies of inflammatory biomarkers multiplexing, as these are vital in monitoring disease progression longitudinally with a single sample [8].

Mechanical stability and reliability in the measurement is also achieved at the sample-sensor interface. Sensor structures built in absorbent layers have been exploited to investigate the dynamics of urine absorption and optimize sensor location [5], and water-soluble, self-destructive films have been exploited to allow tamper-evident and contamination-free sampling [6]. All

these advancements have prompted the design of encapsulated, reusable biosensing modules where the electronic components are preventively isolated against liquid ingress, but sensing electrodes are exposed remotely.

Hardware-wise, recent advances in the design of potentiostat circuits, especially the ultra-low-power and amplifier-less design, have greatly lowered the amount of energy and size, allowing coin-cell-size applications in small-scale diagnostic systems [14], [15]. In the meantime, optical and photonic sensor options still provide low-power options of analyte detection [13].

Last but not least, research on wireless body area networks (WBAN) has considered secure data acquisition and transmission. Mog-based data aggregation schemes that are pairing-free and blockchain-based data collection frameworks provide data integrity and confidentiality without the need to be connected to the internet, which fits into the offline-first operation model of offline rural healthcare [9], [10]. Prior implementations of WBAN have proved the energy-efficient communication of nodes and duty-cycling protocols, and ongoing research on self-powered wearable sensors by IoT has proven sustainable performance and long-range connectivity in the field [12], [17].

Synthesis: The literature reviewed illustrates (i) sensitive and low-cost nanomaterial electrodes and photonic transducers for urinary biomarker detection [1]–[4], [13]; (ii) operational multiplexed microfluidic and smartphone POC systems [7], [8]; (iii) dependable sample-handling and sensing interfaces [5], [6]; (iv) low-power potentiostat architectures to support long battery life [14], [15]; and (v) privacy-preserving data acquisition and communication. Yet, an integrated, reusable capsule that combines all these innovations—including a strong urine-contact geometry, multi-analyte electrochemical sensing, offline data transmission without compromising security, and sustainable functionality—remains an unfulfilled goal. This is the design gap addressed by the proposed Trisense Capsule.

III. WORKING MECHANISM

The Trisense Capsule is a proposed unit, which is an oval-shaped, reusable, diagnostic equipment, suitable to conduct multi-analytic tests on urine in decentralized locations. The capsule contains SPE functionalized with a specialized set of reagents to detect glucose, pH/conductivity, and a third disease-specific biomarker and allows multiplexed electrochemical sensing using a single urine sample chamber. Urine is pumped through a microfluidic inlet into the capsule, and through capillary channels across the electrodes, which has the benefit of maintaining a controlled flow of fluid and contacts the electrodes uniformly.

The electrodes are interfaced with a miniaturized potentiostat circuit comprising of low-power improper operational amplifier, current-mirror biasing, and a digital regulation loop to realize amperometric and potentiometric measurements. An ESP32 microcontroller controls the potentiostat, performing signal conditioning, analog to digital conversion, and noise filtering. To achieve greater precision, temperature-compensation

algorithms are incorporated and reliable readings are achieved even under environmental changes.

Upon the biochemical interaction of the electrode coatings with the urine analytes, the change in current, voltage or impedance is measured and transformed into a quantity that can be measured. The ESP32 codes and sends the diagnostic data over a secure Bluetooth/Wi-Fi local connection to a smartphone it is connected to. Notably, it is designed to operate in an offline-first fashion and does not need any continuous internet, so it is enhanced in rural healthcare.

A rechargeable lithium micro-battery drives the capsule, and is optimized to low-power operation in duty cycles, enabling several test cycles on a single charge. Protective biocompatible housing polymer of an internal hydrophobic barriers prevents leakage and cross-contamination, and the removable sensing module allows reuse of electronics with replaceable electrode strips.

This combined system allows multi-parameter real-time urine analysis in a low-cost, portable and green-friendly form factor. The capsule gives dependable early diagnosis of chronic diseases like diabetes, kidney disorders and urinary infections especially in underserved populations because it removes the need to rely on centralized laboratories or kits with consumables, which are heavy.

IV. SYSTEM IMPLEMENTATION

A. Design and Working Principle

The Trisense Capsule uses a small, integrated system to perform multi-analyte urine testing by integrating controlled fluid management, electrochemical sensing, low-power readout electronics and short-range secure wireless reporting. A capsuled sensing module houses a three-zone SPE with epoxy-separated reagent coatings (glucose oxidase, anti-albumin antibody, zinc oxide). When liquid contacts the device, a prescribed volume of sample is transmitted through the device to the SPE via capillary microchannels, the potentiostat delivers timed bias sequences to each reagent zone, and the embedded ESP32 measures, conditions, and transforms analog responses into diagnostic values. Results are styled, optionally encrypted, and transmitted locally over Wi-Fi access point or BLE to be immediately displayed on a smartphone browser - all the time the device is aggressively duty-cycling and sleeping to save battery life.

B. Mechanical and Electronic Layout

The capsule is an oval (pill) shaped enclosure made in two removable halves: an outer biocompatible polymer shell and an inner service bay with the PCB, battery, and replaceable sensing cartridge. The internal design is stacked: (top) moisture contact / wetness plate and microfluidic inlet, (middle) replaceable SPE cartridge between mechanical guides and epoxy partitions, (bottom) PCB mount with ESP32, LMP91000 (potentiostat frontend), signal amplifier / filter, battery holder and antenna. Electronics are sealed and isolated with hydrophobic barriers. The cartridge can be replaced,

the electrodes changed or cleared or a fresh reagent applied, without breaking open the electronics.

C. Main Components and Roles

- **Coin Cell (LIR2450):** A rechargeable lithium coin cell powers the capsule. It provides compact form factor, sufficient energy density, and supports multiple test cycles.
- **Moisture Detection Plate:** A thin metallic contact connected to a low-power GPIO pin. It detects the presence of urine and wakes the device from deep sleep mode.
- **SPE Cartridge:** Three distinct reagent spots on a carbon SPE; the epoxy walls avoid cross-contamination and allow activation in sequence.
- **Reagents:** (i)GOx: Detects glucose levels. (ii)Anti-Albumin Antibodies: Identifies protein markers linked to kidney health. (iii)ZnO: Enhances electrochemical sensitivity of the electrode.
- **LMP91000 Potentiostat Frontend and Readout Chain:** Provides programmable bias, transimpedance amplification, and analog conditioning for amperometric/ potentiometric measurement.
- **Operational Amplifier:** Boosts weak electrochemical responses, reduces noise, and ensures accurate ADC conversion before processing.
- **ESP32 Microcontroller:** Uses a wake/sleep logic, ADC sample, data processing, temperature compensation, local web server and wireless stack.
- **Enclosure and Seals:** Ingrees O-ring seals around cartridge area, biocompatible polymer housing and hydrophobic coating to prevent ingress.
- **User Interface (Remote):** Local Wi-Fi AP running a basic web interface (192.168.4.1) or BLE GATT profile to display findings without use of the internet or specific apps.
- **Clean Solutions:** (i)PBS: Maintains stable ionic conditions for reactions and calibration. (ii)Distilled Water: Used for dilution, rinsing, and baseline stabilization.

D. Integration and Module Mounting

The SPE cartridge is fixed in a spring-loaded cassette aligning electrodes with PCB test pads; pogo pins are used to offer electrical connection to reduce mechanical wear. The LMP91000 is mounted on a small analog sub-PCB physically isolated of noisy digital circuitry; analog traces are routed with ground pours and guard rings. To measure first contact, a moisture plate is provided that touches the mouth of the capsule, and its signal is sent to an ESP32 hardware interrupt. The lower half has the battery and charging contacts with a mechanical latch to enable a secure removal. Thermal and mechanical tolerances are such that the cartridge may be rinsed (PBS + DI water) and re-inserted without loosening of electrical connections.

E. Operating Mechanism

The Trisense Capsule works by routing urine through microfluidic channels to sensing chambers with screen printed

carbon electrodes covered with selective reagents. The biochemical reactions produced electrochemical signals that are read by a potentiostat-amplifier circuit and digitized by an ESP32 microcontroller for wireless transmission. With disposable sensing chambers and reusable electronics the system offers a cost-effective, handheld solution for multi-disease detection without requiring bulky analyser equipment.

F. Outer (Full Capsule) View of the Trisence Capsule

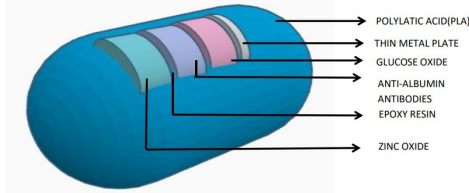


Fig. 1. Outer (Full Capsule) view of the Trisence Capsule

Fig.1 The outer structure shows the capsule's protective and sensing layers. The main body is made of **PLA**, which is lightweight, safe for the body, and biodegradable. A **thin metal plate** provides strength and serves as a base for the sensing layers. The surface coatings include **glucose oxidase** for detecting glucose levels, **anti-albumin antibodies** for identifying protein markers linked to kidney health, and **ZnO**, which boosts the sensitivity of the sensors. Finally, an **epoxy resin coating** seals and protects these sensitive layers from damage, keeping them stable during operation. These outer layers form the first point of contact with the urine sample and enable the device to detect multiple health indicators at once.

G. Systematic (Half-Cut) View of the Trisence Capsule

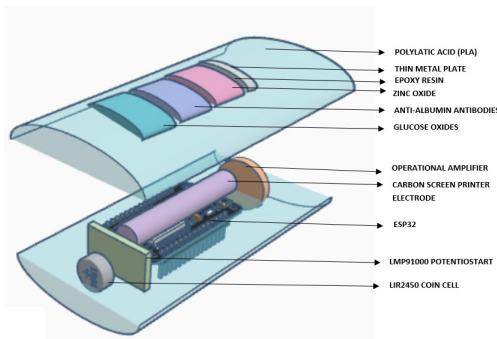


Fig. 2. Systematic (Half-Cut) view of the Trisence Capsule

Fig.2 This view shows how the capsule works from the inside. At its core, a **CSPE** coated with enzyme and antibody layers acts as the main detection surface. When urine interacts with these coatings, tiny electrochemical signals are generated. These signals are very weak, so they are first amplified using an **operational amplifier** and carefully managed by an **LMP91000 potentiostat** to keep the readings stable. The

ESP32 microcontroller then collects and digitizes this data before sending it wirelessly to a mobile device or computer. The system is powered by a small **LIR2450 coin cell battery**, making the device portable and self-sufficient. Together, these components ensure accurate signal detection in a compact design.

V. ANALYTICAL FRAMEWORK AND NUMERICAL COMPUTATION

A. Sensor Response to Analytes

The electrical current generated by each chemical coating in the Trisence Capsule is proportional to the analyte concentration:

$$I_i = k_i \cdot C_i + I_{\text{baseline}}, \quad i = 1, 2, 3$$

where:

- $i = 1 \rightarrow$ Glucose, $i = 2 \rightarrow$ Albumin, $i = 3 \rightarrow$ Zinc Oxide
- I_i = measured current (μA)
- C_i = analyte concentration (mM)
- k_i = sensitivity of the sensor ($\mu A/mM$)
- I_{baseline} = baseline current of blank urine (μA)

B. Normalized Sensor Response (%)

The normalized sensor response is calculated as:

$$R_i = \frac{I_i - I_{\text{baseline}}}{I_{\text{max}} - I_{\text{baseline}}} \times 100$$

where,

- R_i = Normalized sensor response at concentration i
- I_i = Current response at concentration i (measured sensor response)
- I_{baseline} = Baseline current (response in absence of analyte / blank condition)
- I_{max} = Maximum current response (at highest analyte concentration)

C. Signal-to-Noise Ratio (SNR)

The SNR of each analyte is calculated using:

$$\text{SNR}_i = \frac{I_i - I_{\text{baseline}}}{\sigma_{\text{noise}}}$$

Where,

- SNR_i : Signal-to-Noise Ratio of analyte i
- I_i : Measured current response of analyte i (μA)
- I_{baseline} : Baseline current (μA)
- σ_{noise} : Standard deviation of noise (μA)

D. Limit of Detection (LOD)

The limit of detection is defined as:

$$\text{LOD}_i = \frac{3 \cdot \sigma_{\text{noise}}}{k_i}$$

where:

- LOD_i : Limit of Detection for analyte i
- σ_{noise} : Standard deviation of the noise
- k_i : Sensitivity (slope of calibration curve) for analyte i

VI. DATA ACQUISITION AND GRAPHICAL INSIGHTS

TABLE I : NORMALIZED SENSOR RESPONSE(%)

Conc. (mM)	Glu. (%)	Alb. (%)	ZnO (%)	Avg. Response (%)
0.05	0	0	33	11
0.10	0	33	0	11
0.15	0	0	100	33
2.00	40	0	0	13.3
3.00	0	0	0	0
5.00	100	100	0	66.7

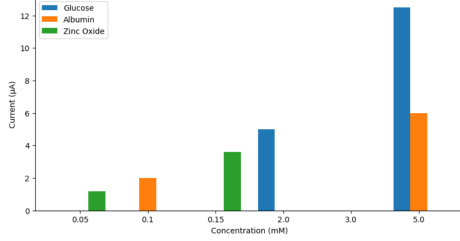


Fig. 3. Sensor Response

TABLE II : SENSOR RESPONSE (A) FOR DIFFERENT CONCENTRATION

Conc. (mM)	Glu.	Alb.	ZnO	Avg. (A)
0.05	0	0	1.2	0.40
0.10	0	2.0	0	0.67
0.15	0	0	3.6	1.20
2.00	5.0	0	0	1.67
3.00	0	0	0	0
5.00	12.5	6.0	0	6.17

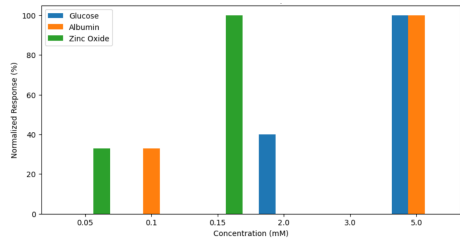


Fig. 4. Normalized Sensor Response

TABLE III : SIGNAL-TO-NOISE RATIO(SNR)

Conc. (mM)	Glu. (SNR)	Alb. (SNR)	ZnO (SNR)	Avg. SNR
0.05	0	0	4	1.33
0.10	0	8	0	2.67
0.15	0	0	12	4
2.00	10	0	0	3.33
3.00	0	0	0	0
5.00	25	24	0	16.33

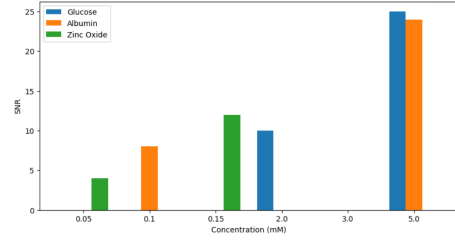


Fig. 5. Signal-to-Noise Ratio(SNR)

TABLE IV : LIMIT OF DETECTION

Analyte	Sensitivity k_i ($\mu A/mM$)	σ_{noise} (μA)	LOD (mM)
Glucose	2.5	0.5	0.6
Albumin	20	0.5	0.075
Zinc Oxide	24	0.5	0.0625

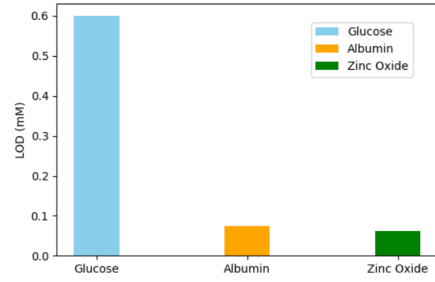


Fig. 6. Limit of Detection(LOD)

VII. SIMULATION RESULT

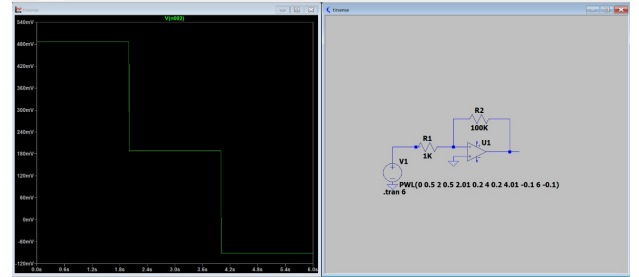


Fig. 7. LTspice transient simulation of the Trisense capsule sensing circuit.

Fig.7 The LTspice transient simulation shows the performance of the proposed Trisense capsule sensing circuit. The design uses an inverting operational amplifier with a $100k\Omega$ feedback resistor and a $1k\Omega$ input resistor, while a piecewise linear voltage source was applied to mimic analyte responses at different concentration levels.

The output waveform clearly displays voltage shifts at around 2 s and 4 s, matching the changes in the input signal. This behavior confirms that the circuit can reliably detect small variations in input and deliver precise millivolt-level outputs. Overall, the results highlight the circuit's ability to provide stable signal amplification, making it a strong candidate for integration into the Trisense capsule for biosensing applications.

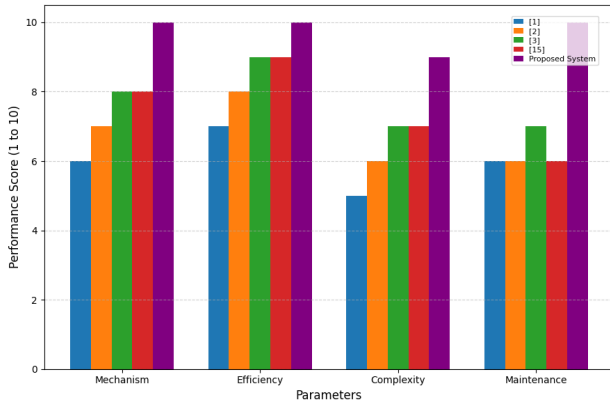


Fig. 8. Comparison of various methods under different parameters

Fig. 8 provides a comparison of the proposed Trisense Capsule to the available reference works [1], [2], [3], and [15] on four parameters, which are; actuation mechanism, energy efficiency, complexity and maintenance. The design is always ranked as the most effective in any category with better responsiveness, optimized power consumption and less construction complexity. It has a better sustainability and depends ability in comparison with previous approaches, which proves its efficiency as a small-size and energy-efficient diagnostic solution.

VIII. CONCLUSION

In this work, a simulation-based design of the Trisense Capsule was developed and validated through analytical and LTspice-based evaluations. The results confirm the reliability of the proposed potentiostat circuit and sensing framework for accurate multi-analyte detection at low power consumption. While the findings are based on theoretical modeling, they provide a strong foundation for prototype realization and wet-lab testing. Future experimental validation will strengthen the system's accuracy and demonstrate its full potential as an affordable, reusable, and eco-friendly diagnostic capsule for multi-disease detection.

IX. FUTURE WORK

The next phase of this study involves the fabrication of a physical prototype of the Trisense Capsule using the described hardware configuration. Wet-lab experiments with synthetic and real urine samples will be conducted to empirically determine parameters such as sensitivity, limit of detection, and stability. These practical evaluations will replace the current theoretical estimates and validate the capsule's suitability for real-world applications in rural and home-based healthcare.

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